

Home Assignment

Clinic Chatbot — Appointment & Visit-Time Suggestion

Overview

Build a simple chatbot for one clinic (working hours 8:00 AM–12:00 PM and 4:00 PM–10:00 PM). The bot should greet the patient, ask if they want a fixed appointment or a walk-in visit, and for walk-ins, use an ML model trained on past visit data to tell them if the clinic is usually busy at that time — suggesting a nearby quieter time if so.

Conversation Flow

- Greet the patient, then ask: “Would you like a fixed appointment, or a walk-in visit?”
- Appointment path: ask date/time, name and email → confirm the booking (guaranteed slot) → send an email copy of the confirmation (mock/log is fine).
- Walk-in path: ask what time they plan to come → send that date/time to the ML model to predict Busy / Normal / Free → if Busy, say so and suggest 1–2 nearby quieter times → if Normal or Free, confirm it's a good time.

Your Task

- Use the sample visit data below to engineer simple features (hour, day-of-week, session).
- Train a small ML model that predicts Busy / Normal / Free for a given date/time.
- Wire this model in as a tool the chatbot calls on the walk-in path, exactly as described above.

Sample Data (2 days of patient visits)

This shows the pattern to expect. Generate more synthetic data in the same style (e.g. 30–60 days) to train your model.

Direct patients	Date	Day	Visit Time
Ramesh Kumar	2026-08-09	Sun	09:05 AM
Priya Suresh	2026-08-09	Sun	09:35 AM
Anitha R	2026-08-09	Sun	10:15 AM
Karthik V	2026-08-09	Sun	10:45 AM
Divya M	2026-08-09	Sun	08:20 AM
Sanjay Patel	2026-08-09	Sun	05:10 PM
Lakshmi N	2026-08-09	Sun	05:40 PM
Arjun S	2026-08-09	Sun	06:15 PM
Meena K	2026-08-09	Sun	07:05 PM
Vijay Raghavan	2026-08-09	Sun	08:30 PM
Kavya Iyer	2026-08-10	Mon	09:05 AM
Suresh Babu	2026-08-10	Mon	09:35 AM
Deepa Nair	2026-08-10	Mon	10:15 AM
Ganesh P	2026-08-10	Mon	10:45 AM
Nithya S	2026-08-10	Mon	08:20 AM
Rahul Menon	2026-08-10	Mon	05:10 PM
Swathi R	2026-08-10	Mon	05:40 PM
Manoj T	2026-08-10	Mon	06:15 PM
Bhavani K	2026-08-10	Mon	07:05 PM
Prakash V	2026-08-10	Mon	08:30 PM

Deliverables

- Working prototype (greeting + both paths) with the ML model wired in.
- Short README: architecture, how LLM and ML model interact, how to run it.